To extend the MIF file to include the upper case, addresses 000h-0FFh were used for lower case letters and addresses 100h-1FFh were used for upper case letters. The first bit represents the case, 0 for lower case, and 1 for upper case. The make codes for both upper and lower case letters are the same. LCD_line variable is created to track the current line. Caps is created to track the current mode. Pair_count is created to count the qualifying pairs, for the top line this is a letter followed by a number and for the bottom line it is a number followed by a letter. LCD_line is initialized to 0, caps set to 0, and char_type is a new variable to track the type of the characters. 0 is for letters, 1 is for numbers, 2 is for a space. In the idle state, shift handling is added. For the left shift, the ps2 code is 8'h12 and this sets caps to 1, making the characters upper case. For the right shift, the ps2 code is 8'h59, and this sets caps to 0, making the characters lower case. The shift bars are also excluded from the character logic to prevent anything from showing up on the display when pressed. The numbers are detected in a different block, and anytime a number is pressed, prev_type is 1. When detecting qualifying pairs, for letter to number pairs, I checked that the previous type was 0, the current type is 1, and the line was 0. If these conditions were met then the pair count would increase. To detect letters, a large condition was created to identify any letter codes. When detecting number to letter pairs, the conditions that need to be met are the previous character is a number (type 1), the current character is a letter (type 0) and the current line is the bottom (1) and if these conditions are met then the count would be increased by 1. If a letter or number is not detected, then the type becomes a space (type 2) and is displayed as a space character. When a new line is created, the count is reset. To connect the shift functionality to the MIF, the case is added to the ps2 code to get the final code. LCD_line gets toggled every time a new line is created. Pair Counts are only shown after all the 16 characters are displayed.